1. Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify:

The security feature/service required, such as confidentiality, integrity, authentication, authorization, availability

The security mechanism used to provide the identified security feature.

The security protocol or technology used to implement the mechanism.

Briefly explain how the mechanism/protocol provides the identified security feature

Application	**Security feature**	**Security Mechanism**	**Protocol/Technology**	**How it provides security**
Gmail	Authentication	Password+MFA	QAuth	Verifies the identity of the user
Online banking	Confidentiality	Encryption	HTTPS/TLS	Protects data transmitted between client and server
Online Payment	Authentication	Pin	HTTPS/TLS	Verifies the identiy of user before transactions
Social media	Authentication	Password	QAuth/HTTPS	Verifies the identiy of user before login
College portal	Authentication and Authorization	Password	HTTPS/TLS	Verifies the user and gives authorization according to their role
Shopping sites	Authentication	Password	HTTPS/TLS	Verifies user account and provides safe transcations
icloud	Confidentiality	Encryption	2 factor authentication/TLS	Verifies user and protects data from unauthorized access

2. For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering from security attacks. Mention the mechanisms and protocols involved.

Application	Security Controls	CIA	Preventing	Detecting	Recovering	Mechanism/ Protocol
Online payment	Encryption, tokenization, MFA, fraud detection	Confidentiality,Integrity,Availability	Protects the user information and does not allow unauthorised access	The fraud detection feature identifies suspicious transactions	Can block accounts which have been compromised	HTTPL/TLS,MFA,tokenization
Social media	Password,Privacy settings, access controls	Confidentiality and integrity	Prevents unauthorised access and protects personal information	While login sends code to registered number which helps detect suspicious login	Password reset can be done using recovery email ids or phone numbers	HTTPS/TLS, OAuth
College Portal	Password,Authentication,Authorization,Backups	Confidentiality,Integrity,Availability	Prevents unauthorised users	Login monitoring and audit calls	Backups and account recov	HTTPS/TLS, MFA

			from accessing and changing confidential information	can detect unusual activities	ery can restore the data and provide access	
Shopping sites	Password,Secure payment	Confidentiality,Integrity,Availability	Protects customer payment information and prevents unauthorised purchasing	Monitors secure transactions and detects suspicious activities	Refunds if any error in transactions and transaction cancellation or account recovery	HTTPS/TLS,tokenization,MFA
icloud	Encryption, 2FA, Backup	Confidentiality,Integrity,Availability	Prevents unauthorised access to user data	Security alert and account monitoring will detect suspicious activities	Backups and account recovery will restore data and access to the account	TLS,end-to-end encryption,2FA